INTERNAL HOM INTO A BOUNDED INJECTIVE COMPLEX HAS FINITE COHOMOLOGY MODULES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The truncation distinguished triangle for a complex. Invoked at the start of the counted proof (source lines 2798-2802) as a cross-chapter pointer. Splits the complex into a bounded-above part and a tail.
- The statement that the internal hom of a bounded-above coherent complex into a bounded complex with finite injective dimension has finite cohomology. Invoked at the end of the counted proof (source lines 2809-2812) as a cross-chapter pointer. Gives finiteness of the cohomology of the truncated part.

Source reference. Original reference derived-remark-truncation-distinguished-triangle: sources/stacks-dualizing/pinned-original-source.tex, source lines 2803–2803. Navigation pointer only: the counted text refers to the Stacks item derived-remark-truncation-distinguished-triangle, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference perfect-lemma-coherent-internal-hom: sources/stacks-dualizing/pinned-original-source.tex, source lines 2810–2810. Navigation pointer only: the counted text refers to the Stacks item perfect-lemma-coherent-internal-hom, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:dualizing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let A be a Noetherian ring. Let $K, L \in D_{Coh}(A)$ and assume L has finite injective dimension. Then $R\mathrm{Hom}_A(K, L)$ is in $D_{Coh}(A)$.*

Proof. Pick an integer n and consider the distinguished triangle

$$\tau_{\leq n}K \to K \to \tau_{\geq n+1}K \to \tau_{\leq n}K[1]$$

see Derived Categories, Remark derived-remark-truncation-distinguished-triangle. Since L has finite injective dimension we see that $R\mathrm{Hom}_A(\tau_{\geq n+1}K, L)$ has vanishing cohomology in degrees $\geq c - n$ for some constant c. Hence, given i, we see that $\mathrm{Ext}^i_A(K, L) \to \mathrm{Ext}^i_A(\tau_{\leq n}K, L)$ is an isomorphism for some $n \gg -i$. By Derived Categories of Schemes, Lemma perfect-lemma-coherent-internal-hom applied to $\tau_{\leq n}K$ and L we see conclude that $\mathrm{Ext}^i_A(K, L)$ is a finite A-module for all i. Hence $R\mathrm{Hom}_A(K, L)$ is indeed an object of $D_{Coh}(A)$. □